

Literature Survey

Sensitive Data Detection using an Agentic-AI Framework and Machine Learning Models

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1. Introduction

The rapid digitisation of personal, financial, and government records has intensified the challenge of protecting Personally Identifiable Information (PII) embedded in unstructured text. Conventional rule-based and pattern-matching approaches to PII detection have proven brittle across heterogeneous document types and increasingly fail to capture context-dependent sensitive content. Emerging research addresses this gap through three complementary directions: (1) transformer-based NLP models that contextually identify and mask PII, (2) privacy-preserving anonymisation frameworks that balance data utility with de-identification guarantees, and (3) adversarial evaluation techniques that stress-test the robustness of both detection and masking pipelines. The papers reviewed below span each of these directions and collectively motivate the agentic, multi-stage system proposed in this project.

2. Review of Related Works

2.1 NLP-Based PII Detection Using Transformer Models

Mainetti and Elia (2025) addressed the detection of PII in unstructured textual data, motivated by escalating data-breach costs and the tightening requirements of the General

Data Protection Regulation (GDPR). They compiled a training corpus by combining approximately 43,946 English records drawn from the publicly available pii-masking-200k dataset — which spans 54 PII classes across 229 use-case domains — with 42,998 non-PII records sourced from the GenericSentiment dataset on Kaggle, yielding a perfectly balanced binary corpus. For testing, they assembled a separate multi-source dataset comprising records from the pii-masking-43k benchmark, the Dialog Dataset of chatbot conversations, 10,000 IMDB movie reviews confirmed PII-free, and 1,000 GPT-4-generated sentences. Mainetti and Elia retrained the pre-trained BERT_LARGE model using three strategies — a standard training-validation-test (TVT) split allocating 80% for training and 10% each for validation and testing, five-fold cross-validation, and stratified five-fold cross-validation — implemented in Python 3.11 with PyTorch and TensorFlow on an NVIDIA RTX 4090 GPU. The software pipeline was decomposed into four classes — PiiDataLoader, DataHandler, PiiDataset, and DetectionModelTrainer — with the Hugging Face Transformers tokeniser employed for sub-word tokenisation and a pre-trained Named Entity Recognition model used to label PII spans. After three training epochs with AdamW optimisation at a learning rate of 5×10^{-5} , the TVT model achieved 99.558% accuracy, 99.564% precision, 99.558% recall, and an F1-score of 99.559%, surpassing the best-performing comparison model (Mistral 7B) by 7.47 percentage points in accuracy. The study is limited by its reliance on a single pre-trained architecture, as alternative encoders such as RoBERTa and DeBERTa are not explored, and the testing datasets differ from those used in comparative works, making direct benchmark comparisons difficult.

2.2 Privacy-Preserving Anonymisation and Data Utility

Gajavalli (2025) investigated whether incorporating human expert input into the k-anonymisation process could improve the balance between data utility and privacy, a problem of direct relevance to Privacy-Aware Machine Learning (PaML) under GDPR. He used the UCI Adult dataset, originally derived from 1994 US Census Bureau records, retaining key attributes including age, race, gender, ZIP code, education-num, and income after excluding skewed or redundant fields such as capital-gain, fnlwgt, and

education. Gajavalli adopted the SaNGreeA greedy clustering algorithm with $O(n^2)$ complexity to generate equivalence classes and minimise General Information Loss (GIL); he produced 775 anonymised dataset variants by sweeping the anonymisation k -factor from 5 to 200 on a 3,000-record subset, and validated classifier utility using linear SVC, logistic regression, gradient boosting, and random forest across three prediction targets — income, education level, and marital status. The interactive machine learning (iML) condition allowed human participants to adjust quasi-identifier attribute weights in real time through a browser-based interface before each anonymisation step, while equal-weighting and user-defined-bias conditions served as baselines. Results showed that human-guided weighting outperformed both baselines for the marital-status target, while performance for income and education level was inconsistent across classifiers and k values; notably, even $k = 200$ occasionally preserved comparable classifier F1-scores. The study is limited by its reliance on a simplified $O(n^2)$ algorithm applied to only 500 rows for the interactive condition, which likely accelerates over-generalisation and prevents conclusions about scalability to realistic document volumes.

2.3 Adversarial Robustness of NLP Models

Madrueno, Fernandez-Isabel, Fernandez, and Martin de Diego (2025) proposed a novel black-box adversarial attack method for NLP classifiers that combines sentence-, word-, and character-level text perturbations generated entirely through Large Language Model instruction prompts. They used two binary sentiment classification datasets: SST-2, consisting of 872 single-sentence movie-review excerpts from Rotten Tomatoes, and IMDB, comprising 500 full-length reviews from the Internet Movie Database. Madrueno, Fernandez-Isabel, Fernandez, and Martin de Diego employed GPT-4o mini to generate four types of LLM-driven perturbations — sentence paraphrases, word-level typos, synonym replacements, and neutral word insertions — applied within a two-stage pipeline: first, each sentence of the original input was paraphrased using an LLM instruction prompt; second, the 40% most influential words identified via a Word Importance Ranking (WIR) were further perturbed with typos, synonyms, or surrounding word insertions, evaluated in decreasing order of impact on victim model confidence

scores. The method was validated against the instruct versions of four open-weight LLMs — Gemma 2 9B, Llama 3.1 8B, Qwen 2.5 7B, and Yi 1.5 6B — using Attack Success Rate (ASR) and semantic preservation measured independently by GPT-4o. The proposed all-perturbation configuration achieved ASR scores of 0.90 to 0.96 on SST-2 and 0.87 to 0.98 on IMDB across all victim models, outperforming every single-perturbation baseline by 29 to 85%, while maintaining semantic preservation above 0.94 across all experimental conditions. The method is limited by its substantial and unpredictable computational cost, as generating and evaluating all candidate perturbations for each vulnerable word requires a large number of LLM queries, making worst-case adversarial example generation extremely resource-intensive.

3. Summary Comparative Table

Table 1 summarises the key properties of each reviewed study.

Author(s)	Year	Methods / Models Used	Key Contribution	Study Limitation (present tense)
Mainetti & Elia	2025	BERT_LARGE; Hugging Face Transformers tokeniser + NER; TVT split, k-fold, stratified k-fold; AdamW optimiser (lr = 5e-5); Python 3.11 / PyTorch / TensorFlow	Re-trained BERT_LARGE on a balanced 87k-record corpus (pii-masking-200k + GenericSentiment) covering 54 PII classes; TVT model achieved 99.558% accuracy, 99.559% F1, surpassing Mistral 7B by 7.47 pp	The approach is limited to a single architecture (BERT); alternative encoders (RoBERTa, DeBERTa) are not evaluated, and differing test sets complicate cross-study comparison
Gajavalli	2025	SaN GreeA greedy k-anonymisation ($O(n^2)$); GIL metric; linear SVC, logistic regression, gradient boosting, random forest; iML browser-based interface	Proposed human-guided interactive k-anonymisation showing domain-expert quasi-identifier weighting can preserve classifier utility (F1) better than automated equal-weighting, particularly for marital-status prediction on the UCI Adult dataset	The framework is limited by its $O(n^2)$ algorithm and a 500-row interactive subset, preventing scalability and causing premature over-generalisation
Madru��o, Fernandez-Isabel, Fernandez & Martin de Diego	2025	GPT-4o mini (attacker LLM); Gemma 2 9B, Llama 3.1 8B, Qwen 2.5 7B, Yi 1.5 6B (victim LLMs); Word Importance Ranking; multi-	Proposed a two-stage LLM-driven black-box adversarial attack combining paraphrases, typos, synonyms, and word insertions; achieved ASR of 0.90-0.98 across four victim LLMs on SST-2 and IMDB, outperforming single-perturbation baselines by 29-85% with semantic preservation above 0.94	The method is limited by high and unpredictable computational cost arising from the large number of LLM queries needed to exhaustively generate and score all candidate perturbations in worst-case scenarios

Author(s)	Year	Methods / Models Used	Key Contribution	Study Limitation (present tense)
		level LLM perturbations (paraphrase, typo, synonym, insertion)		

4. Synthesis and Research Gap

The three works reviewed above collectively delineate the major technical axes on which robust PII protection in unstructured documents must be built. Mainetti and Elia (2025) demonstrated that a contextually rich transformer model, when re-trained on a sufficiently large and balanced PII corpus, can achieve near-perfect token-level detection accuracy. However, their pipeline treats detection as an isolated stage and relies on a single architecture, leaving open the questions of how masking strategies should be chosen after detection, how re-identification risk should be quantified following masking, and whether the masked document retains adequate downstream utility.

Gajavalli (2025) confronted the utility-privacy trade-off directly by introducing human-in-the-loop guidance into the k-anonymisation process. While his results confirm that domain knowledge can preserve task-relevant quasi-identifiers better than fully automated heuristics, the study also highlighted that human judgment is inconsistent across classification targets and that greedy anonymisation algorithms cannot scale to the document volumes encountered in real-world PII processing pipelines.

Madrueno, Fernandez-Isabel, Fernandez, and Martin de Diego (2025) shifted the evaluation lens from detection accuracy to adversarial robustness, demonstrating that semantics-preserving LLM-generated text perturbations can systematically deceive even well-performing NLP classifiers. This finding is directly relevant to the proposed project: a PII detection and masking pipeline that is not evaluated against adversarial inputs may produce a false sense of privacy compliance, since an attacker or a document that naturally diverges from training distributions could evade detection.

Taken together, these studies reveal a common and critical gap: no existing system integrates contextual PII detection, adaptive masking strategy selection, utility evaluation, and adversarial privacy analysis within a single coherent pipeline. The proposed agentic architecture addresses precisely this gap by decomposing PII handling across specialised autonomous agents — each responsible for one of these concerns — and enabling coordinated, holistic reasoning across all stages of the privacy-preservation workflow for unstructured documents.